

FRANCESCO BARILE

CONTACT INFORMATION

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ACADEMIC POSITIONS

University of Milano-Bicocca, Milan, Italy

• **Postdoctoral Research Fellow**

Mar 2025 - present

(*Assegno di Ricerca* pursuant to Article 22 of Italian Law No. 240/2010)

Department of Economics, Management and Statistics

Project: Joint Bayesian Learning of Related Clinical Studies

Funding: Fondazione Cariplo

Project PI: Prof. Bernardo Nipoti, University of Milano-Bicocca

Partner unit's coordinator: Prof. Francesco C. Stingo, University of Florence

Project members: Prof. Matteo Borrotti, University of Milano-Bicocca;

Dr. Giovanni Poli, University of Florence

WORK EXPERIENCE

• **Junior Data Scientist**

Apr 2019 - Sep 2020

CRIF S.p.A.

• **Degree Program Tutor**

Feb - Jul 2018

Department of Economics, Statistics and Finance, University of Calabria

• **Degree Program Tutor**

Apr - Oct 2017

Department of Economics, Statistics and Finance, University of Calabria

EDUCATION

University of Milano-Bicocca, Milan, Italy

• **Ph.D. in Statistics**

2020 - 2024

Department of Economics, Management and Statistics

Defense date: Feb 14th, 2025

Thesis title: "Advances in Bayesian Methods:

Final evaluation: Excellent

Nonparametric Network Modeling and Scalable Computations"

Supervisor: Prof. Bernardo Nipoti

University of Calabria, Rende, Italy

• **M.Sc. in Finance and Insurance** (English)

2016 - 2019

Department of Economics, Statistics and Finance

Thesis title: "Estimating Stochastic Models for commodity futures"

110/110 cum laude

Supervisor: Prof. Arturo Leccadito

• **B.Sc. in Statistics for Business and Insurance**

2012 - 2015

Department of Economics, Statistics and Finance

Thesis title: "Wilcoxon test: applications in the medical-pharmaceutical field"

110/110 cum laude

Supervisor: Prof. Agostino Tarsitano

RESEARCH VISITS

•Visiting Researcher

Nov 2023 - Aug 2024

Lancaster University, Lancaster, United Kingdom
Department of Mathematics and Statistics
Supervisor: Prof. Christopher Nemeth

•Visiting Graduate Student (Research Only)

Aug - Dec 2018

University of Western Ontario, London, Canada
Department of Statistical and Actuarial Sciences
Supervisor: Prof. Rogemar Mamon
Research topic: Statistical modeling of lumber price dynamics

RESEARCH INTERESTS

Bayesian nonparametrics, Network modeling, Computational statistics

SCHOLARSHIPS AND AWARDS

•Junior Travel Support Award

2026

ISBA 2026, Nagoya, Japan

•Junior Travel Support Award

2025

BayesComp 2025, Singapore

•Exchange Extra-UE Funded Mobility Programme Award (10 months)

2023

University of Milano-Bicocca

•Ph.D. Scholarship Award (4 years)

2020

University of Milano-Bicocca

•Overseas Research Scholarship Award (4 months)

2018

University of Calabria

PUBLICATIONS

REFERRED JOURNAL ARTICLES

REFERENCES

- [1] Barile, F., Lunagómez, S., & Nipoti, B. (2025). Bayesian nonparametric modeling of heterogeneous populations of networks. *Bayesian Analysis*. doi: 10.1214/26-BA1588. [\[link\]](#)

REFERRED CONFERENCE PROCEEDINGS

REFERENCES

- [1] Barile, F., Lunagómez, S., & Nipoti, B. (2023). Flexible modelling of heterogeneous populations of networks: a Bayesian nonparametric approach. *Book of abstracts and short papers of CLADAG 2023*. [\[link\]](#)
- [2] Barile, F., Lunagómez, S., & Nipoti, B. (2025). Clustering multiple network data with an application to the EU food trade market. *Book of Short Papers of SIS 2024*. [\[link\]](#)
- [3] Barile, F., Dallari, S., Grün, B., & Pacifici, C. (2025). Assessing neuron response to external stimuli with a data-driven procedure for spike train extraction and GAMLSS regressions. *Book of Proceedings of the 2022 Data Research Camp*. [\[link\]](#)

MANUSCRIPT UNDER REVIEW

REFERENCES

- [1] Barile, F. & Nemeth, C. (2025). Control variate-based stochastic sampling from the probability simplex. *Under review. ArXiv: [\[link\]](#). Code: [\[link\]](#)*
- [2] Barile, F., Capozio, S. & Nipoti, B. (2026). Efficient Bayesian inference for multiple network data. *Under review. Code: [\[link\]](#)*

MANUSCRIPT IN PREPARATION

REFERENCES

- [1] Barile, F., Borrotti, M., Nipoti, B., Poli, G. & Stingo, F. C. (2026+). Data Integration and Missing Imputation for Joint Bayesian Learning under Multiple Nonignorable Missing Mechanism. *In preparation.*
- [2] Barile, F., Nieto-Barajas, L.E., & Nipoti, B. (2026+). A Network of Dependent Dirichlet Processes. *In preparation.*
- [3] Barile, F., Lunagómez, S., & Nemeth, C. (2026+). Covariate-dependent modeling of hypergraph data. *In preparation.*

PH.D. THESIS

REFERENCES

- [1] Barile, F. (2025). Advances in Bayesian Methods: Nonparametric Network Modeling and Scalable Computations. Ph.D. thesis, XXXVI cycle [\[link\]](#)

PRESENTATIONS AND SEMINARS

INVITED PRESENTATIONS

- CFE-CMStatistics 2026, HTW Berlin, University of Applied Sciences, Berlin, Germany Dec 12-14, 2026
- StaTalk 2025, University of Milano-Bicocca, Milan, Italy Jun 13-14, 2025
- ECOSTA 2023, Waseda University, Tokyo, Japan Aug 1-3, 2023

CONTRIBUTED PRESENTATIONS

- BAYSM 2023 Nov 13-17, 2023

POSTERS

- ISBA 2026, Nagoya, Japan Jun 28 - Jul 3, 2026
- Bayes Comp 2025, Singapore Jun 16-20, 2025
- SISBayes, Padova, Italy Sep 4-5, 2025
- Complex Graphical Models for Biological Network Science, Padova, Italy Sep 4, 2024
- ISBA 2024, Venice, Italy Jul 1-7, 2024
- IMS ICSDS 2022, Florence, Italy Dec 13-16, 2022

SEMINARS

- Seminar in the Computational Statistics and Machine Learning series Feb 2024
School of Mathematical Sciences, Lancaster University, Lancaster, UK

WORKSHOPS

- Data Research Camp 2022, VIU, San Servolo Island, Venice, Italy Jul 12-15, 2022

TEACHING

Lecturer

- Inferential Statistics [ENG] 2026/2027
(Master's degree in International Economics and Master's degree in Global Management)
Department of Economics, Management and Statistics, University of Milano-Bicocca

Adjunct Lecturer

- Industrial Economics and Dynamics [ITA] 2021/2022
(Master's degree in Statistical and Economic Sciences)
Department of Economics, Management and Statistics, University of Milano-Bicocca

Teaching Assistant

- Inferential Statistics [ENG] (Master's degree in International Economics) 2025/2026
Department of Economics, Management and Statistics, University of Milano-Bicocca
- Inferential Statistics [ENG] (Master's degree in International Economics) 2024/2025
Department of Economics, Management and Statistics, University of Milano-Bicocca
- Advanced Statistics [ENG] (Master's degree in Marketing and Global Markets) 2024/2025
Department of Economics, Management and Statistics, University of Milano-Bicocca
- Inferential Statistics [ENG] (Master's degree in International Economics) 2023/2024
Department of Economics, Management and Statistics, University of Milano-Bicocca
- Advanced Statistics [ENG] (Master's degree in Marketing and Global Markets) 2023/2024
Department of Economics, Management and Statistics, University of Milano-Bicocca
- Inferential Statistics [ENG] (Master's degree in International Economics) 2022/2023
Department of Economics, Management and Statistics, University of Milano-Bicocca

SERVICE TO PROFESSION (MANCA ISCRIZIONI A SOCIETA' ES. SIS)

ORGANIZATION OF SCIENTIFIC EVENTS

Organizer and chair of the Invited Session "Recent advances in Bayesian variable selection and shrinkage" of CFE-CMStatistics 2026 [\[link\]](#)

REFEREE SERVICE

Computational Statistics and Data Analysis; Communications in Statistics - Simulation and Computation; IEEE Journal of Biomedical and Health Informatics.

THESIS SUPERVISION

Co-supervision of a Master's thesis, Master's Degree in Statistical and Economic Sciences, Department of Economics, Management and Statistics, University of Milano-Bicocca

COMPUTER SKILLS

Programming languages	R, C++, Python (basic), SQL, Java, $\text{\LaTeX}$
Software	SAS, Stata RStudio: • C++ based packages creation; • parallel programming; • create dynamic reports and reproducible research using $\text{\LaTeX}$; • project version control via GitHub.

LANGUAGES

Italian Mother tongue
English Fluent
French Basic

The statements contained in this curriculum vitae are made pursuant to Articles 46 and 47 of Presidential Decree No. 445/2000.

Milan, September 23, 2026